Code No: 117EE **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, July/August - 2022
LINUX PROGRAMMING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Answer Any Five Questions
All Questions carry equal marks
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1.a) Write a detail note on shell meta characters and shell commands available in Bash.
b) Explain briefly about various backup utilities and disk utilities with few options. [8+7]

2.a) List and explain various string and mathematical functions available in awk with examples for each.
b) Write an awk script to print the first *n* prime numbers into a file. [8+7]

3.a) What is meant by file mode creation? How to set it for a process?
b) Demonstrate various file mode constants available for chmod function. [6+9]

4.a) Write about stat( ), fstat( ) and lstat( ) system calls. Explain how these three differ with each other.
b) Demonstrate with a sample code, various system calls used to copy the content of one file to another file. [8+7]

5.a) Differentiate between fork() and exec() system calls.
b) Describe the UNIX kernel support for a process. Show the related data structures with a neat diagram. [8+7]

6.a) What are the signals that are not ignored or blocked? Explain the reason behind it with an example.
b) What is the need of exec() system calls? Explain with an illustrative example. [8+7]

7.a) Compare the IPC functionality provided by Pipes, FIFOs and Message Queues.
b) Explain various fields of ipc_perm structure. [6+9]

8. Explain briefly about the following socket APIs with clear syntax and example: [15]
a) socket() b) bind() c) listen() d) accept()

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